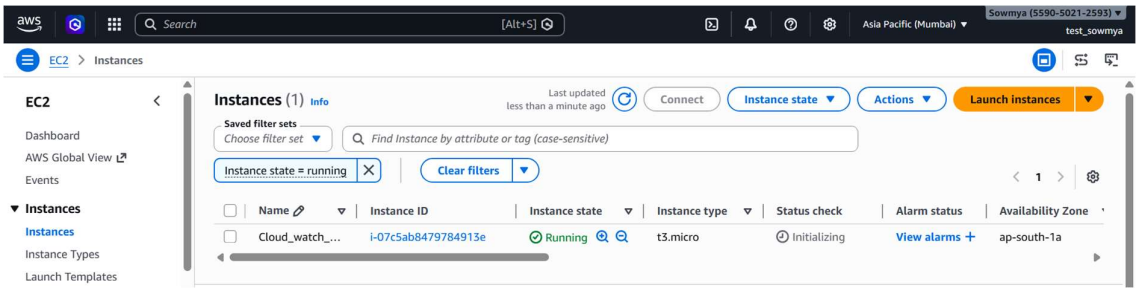
Demonstrating Default and Custom EC2 Metrics in CloudWatch

Step 1: Attach Cloud watch Permission to your IAM user and Login with your IAM user account in AWS console.

Step 2: Search for Ec2 and create an ec2 instance for cloud watch demo.

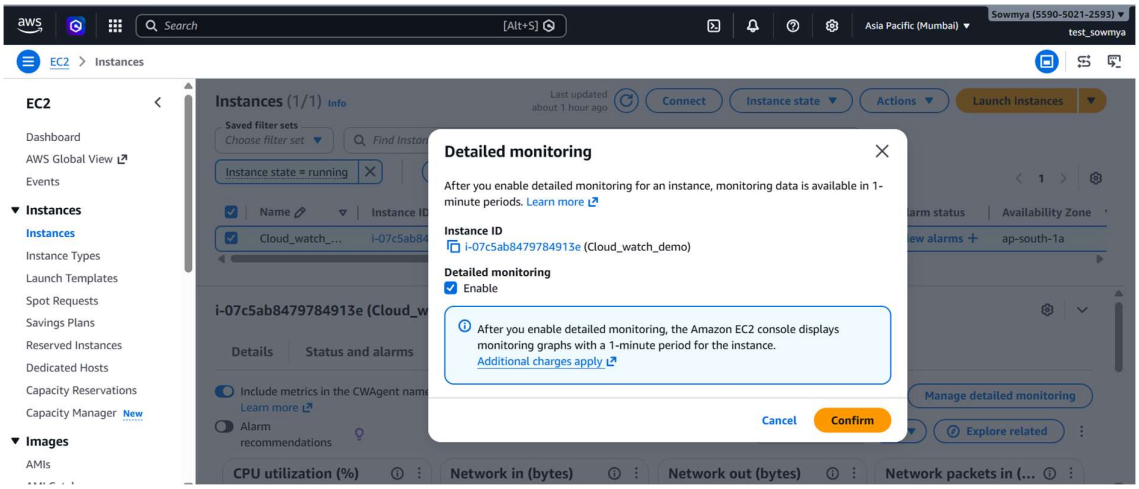


Step 3: To connect to ec2 from windows i.e WSL follow below steps

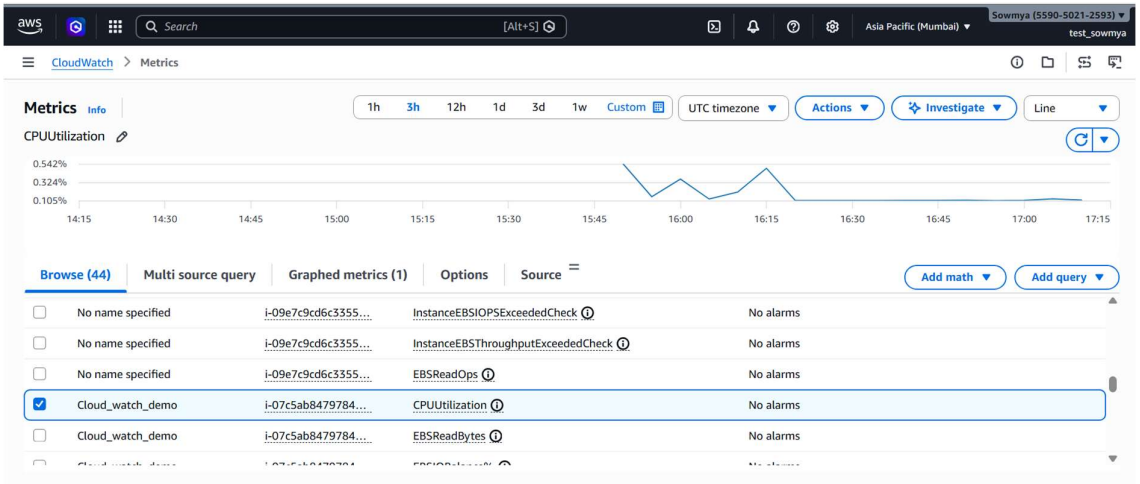
1. Open Command Prompt and type wsl to start Windows Subsystem for Linux.
2. Switch to the home directory using cd ~.
3. Move the key pair file from Windows to WSL using:
   mv /mnt/c/Users/<your-username>/Downloads/<keypairfilename>.pem ~/
4. Set proper permissions for the key file using:
   chmod 400 ~/<keypairfilename>.pem
5. Connect to the EC2 instance using:
   ssh -i ~/<keypairfilename>.pem <username>@<public-ip-address>

Step 4: By default, Ec2 instance sends metrics every 5 min but if you want to monitor in detail. Go to Ec2->select the instance you created->Monitoring->Click on Manage detailed monitoring.

Enable detailed monitoring then it will send metric every 1 min.

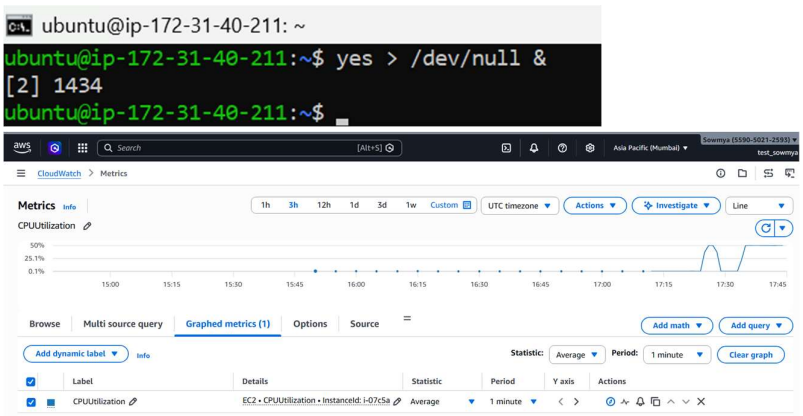


Step 5: Search cloud watch->Metrics->All metrics->Ec2->Per instance Metrics->select your instance and observe the spike level.

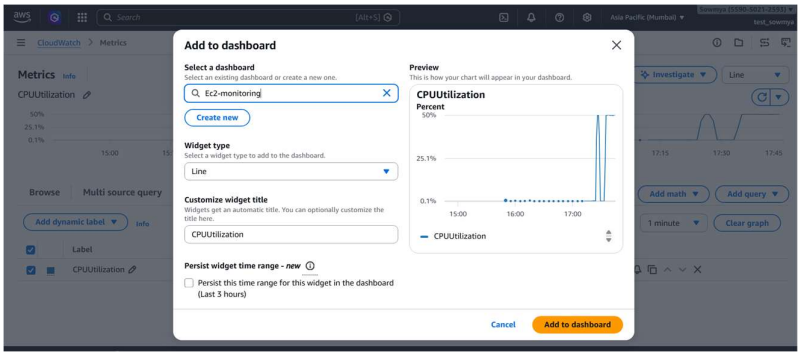


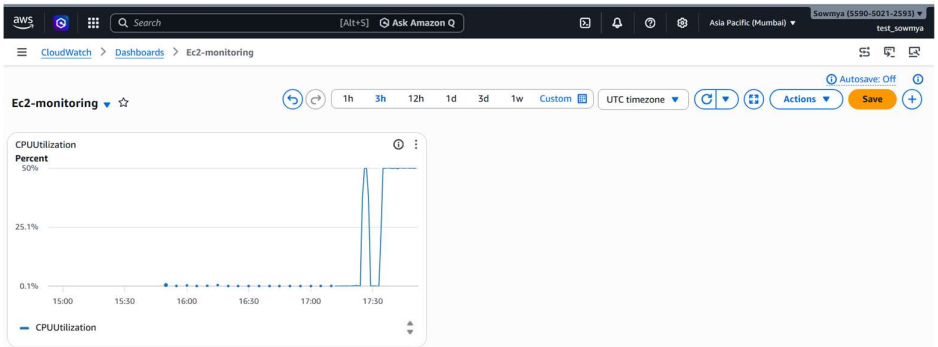
Step 6: Observe the spike by selecting period as 1 min and then to increase the cpu load using command below then observe the spike

`yes > /dev/null &` →Runs an infinite CPU-consuming process in the background, but don't display anything



Step 7: You can also add to dashboard . Select->Actions->Add to dashboard. Here create a dashboard of ec2-monitoring and click on add to dashboard





To stop the cpu consuming process use pkill yes

```
ubuntu@ip-172-31-40-211:~$ yes > /dev/null &
[2] 1434
ubuntu@ip-172-31-40-211:~$ pkill yes
[2]-  Terminated              yes > /dev/null
ubuntu@ip-172-31-40-211:~$
```

We can use **CloudWatch Alarms** to receive notifications when CPU utilization exceeds a defined threshold. These alarms must be configured by specifying the metric, threshold value, evaluation period, and notification action.

Custom Metric-> A custom metric is a user-defined metric that you push to Amazon CloudWatch using the AWS CLI, SDK, or agent.

Let's send a custom metric called memory utilization from CPU

Step 1: Install AWS CLI using below command

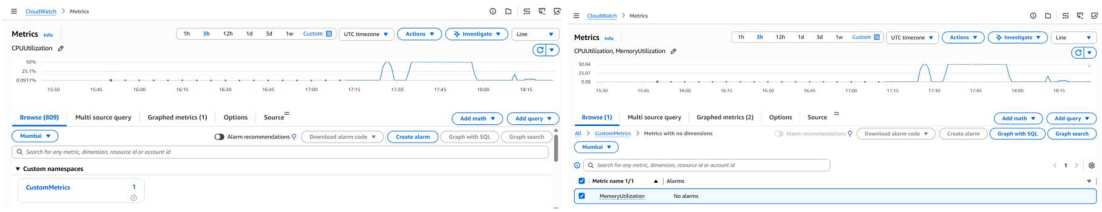
sudo apt update

sudo apt install awscli -y or use sudo snap install aws-cli --classic

Step 2: Create a IAM role for Ec2 to access Cloud watch with Cloudwatchputmetricdata permission and the run below command

```
aws cloudwatch put-metric-data \
 --namespace "CustomMetrics" \
 --metric-name "MemoryUtilization" \
 --value 45 \
 --region ap-south-1
```

You will be able to see custom metric created in your cloudwatch



Step 3: You can also send dynamic value. Use below commands

free | grep Mem | awk '{print ($3/$2)*100}' -shows memory usage and calculates the percentage. Here the formula used is (used memory / total memory) * 100

MEM=$(free | grep Mem | awk '{print ($3/$2)*100}') –stores value in variable MEM.

aws cloudwatch put-metric-data \

--namespace "CustomMetrics" \

--metric-name "MemoryUtilization" \

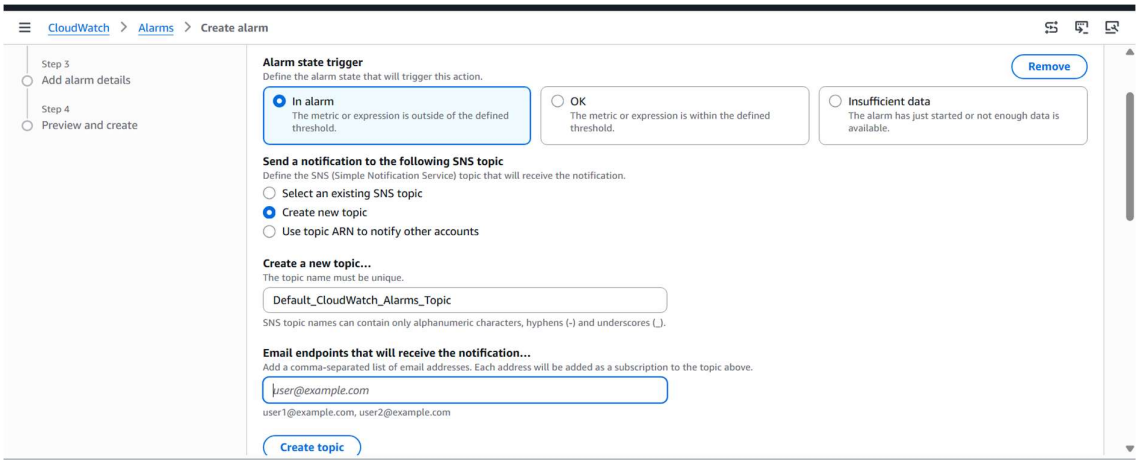
--value $MEM \

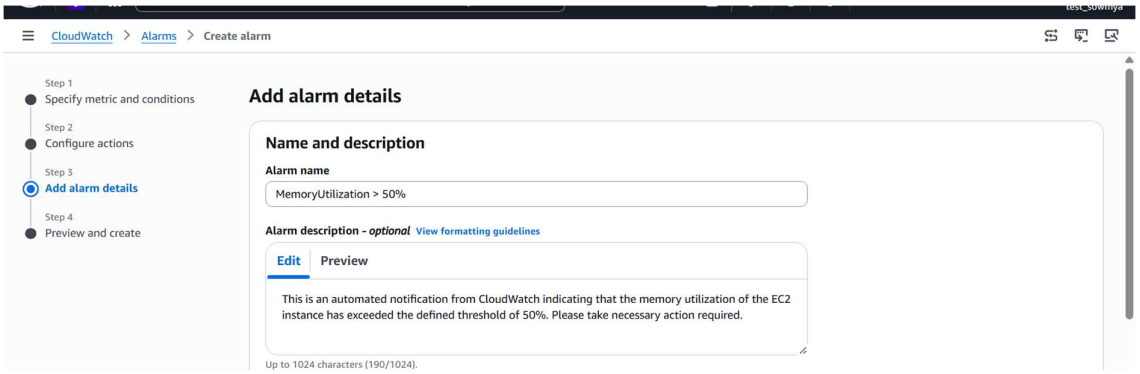
--region ap-south-1 This is to send data to cloud watch

watch -n 60 'MEM=$(free | grep Mem | awk "{print (\$3/\$2)*100}"); aws cloudwatch put-metric-data --namespace CustomMetrics --metric-name MemoryUtilization --value $MEM --region ap-south-1' –It runs every 60sec and calculates memory and sends metric to cloud watch

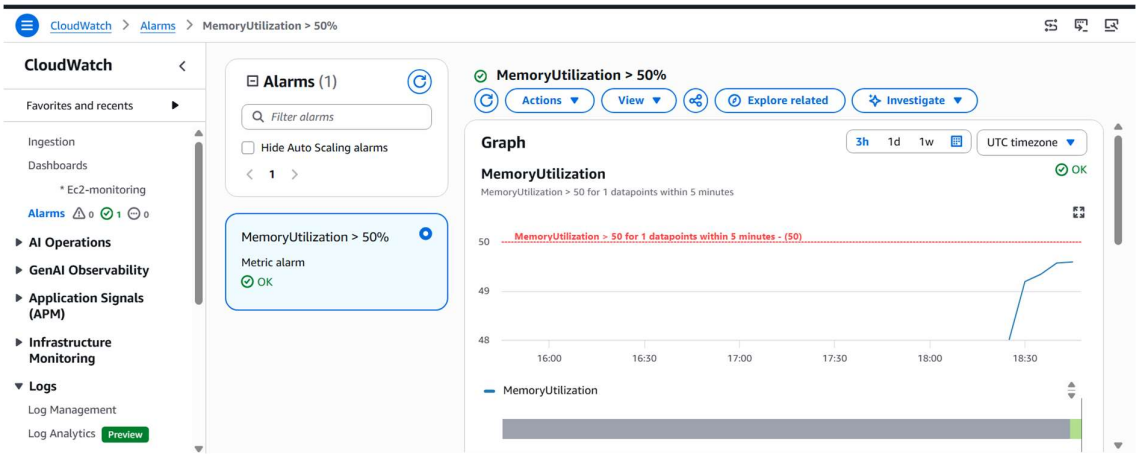
Step 4: Lets create an alarm to get notifications when memory utilizations>50%

Go to Cloud watch->alarm->create alarm->select memory utilization metric->set threshold value 50->create topic, add email->





Step 5: You will get a mail notification. Click on confirm subscription. Now check the alarm status. Once the threshold value reaches 50 you will get a mail notification.



ALARM: "MemoryUtilization > 50%" in Asia Pacific (Mumbai) Inbox ×

AWS Notifications <no-reply@sns.amazonaws.com> 12:24 AM (2 minutes ago)

to me

You are receiving this email because your Amazon CloudWatch Alarm "MemoryUtilization > 50%" in the Asia Pacific (Mumbai) region has entered the ALARM state, because "Threshold Crossed: 1 out of the last 1 datapoints [60.398174999999995 (25/04/26 18:49:00)] was greater than the threshold (50.0) (minimum 1 datapoint for OK -> ALARM transition)." at "Saturday 25 April, 2026 18:54:02 UTC".

View this alarm in the AWS Management Console:
https://ap-south-1.console.aws.amazon.com/cloudwatch/deeplink.js?region=ap-south-1#alarmsV2:alarm/MemoryUtilization%20%3E%2050%25

Alarm Details:
- Name: MemoryUtilization > 50%
- Description: This is an automated notification from CloudWatch indicating that the memory utilization of the EC2 instance has exceeded the defined threshold of 50%. Please take necessary action required.
- State Change: OK -> ALARM
- Reason for State Change: Threshold Crossed: 1 out of the last 1 datapoints [60.398174999999995 (25/04/26 18:49:00)] was greater than the threshold (50.0) (minimum 1 datapoint for OK -> ALARM transition).

Successfully Demonstrated Default and Custom metrics!!